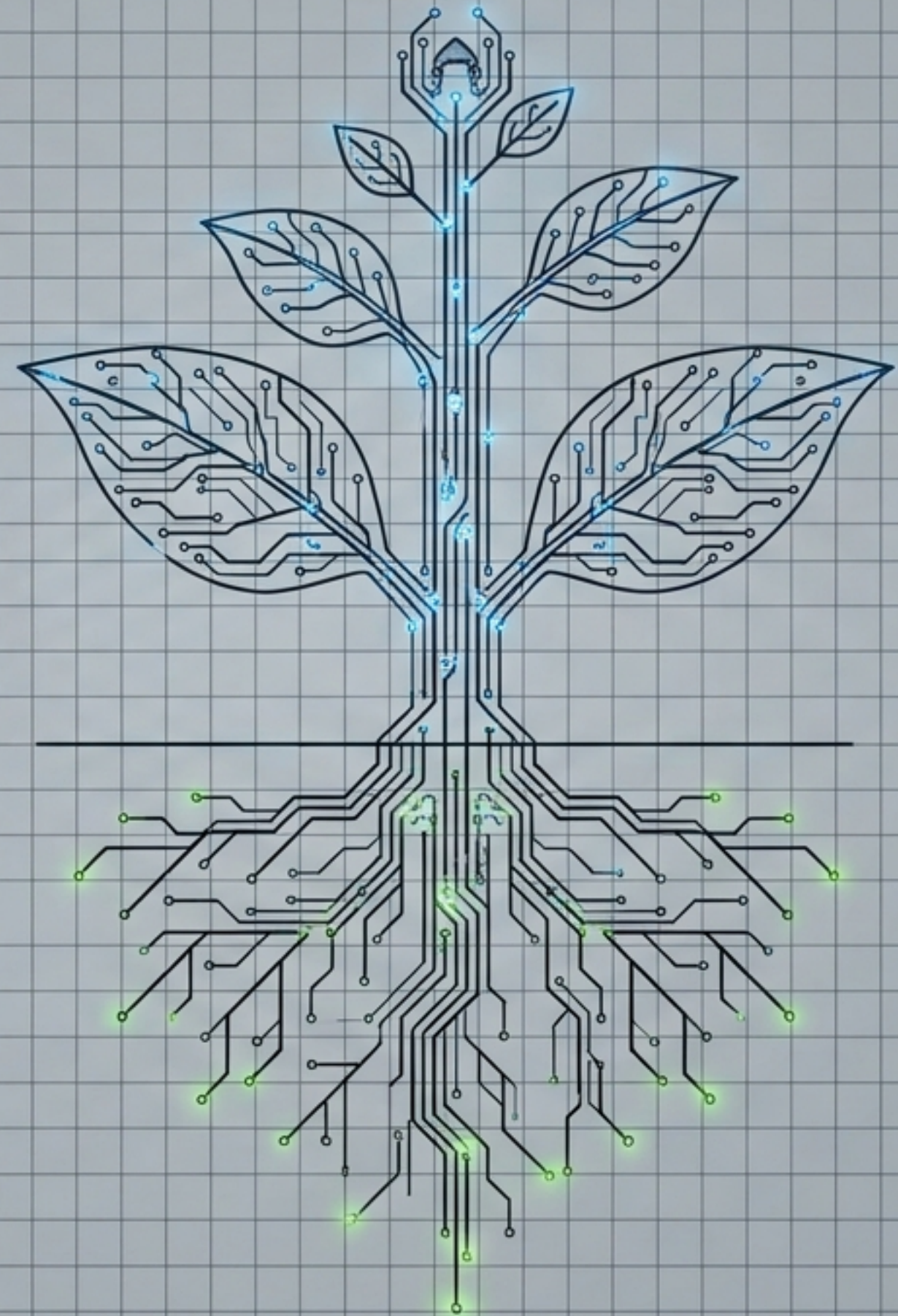


MODULE 1

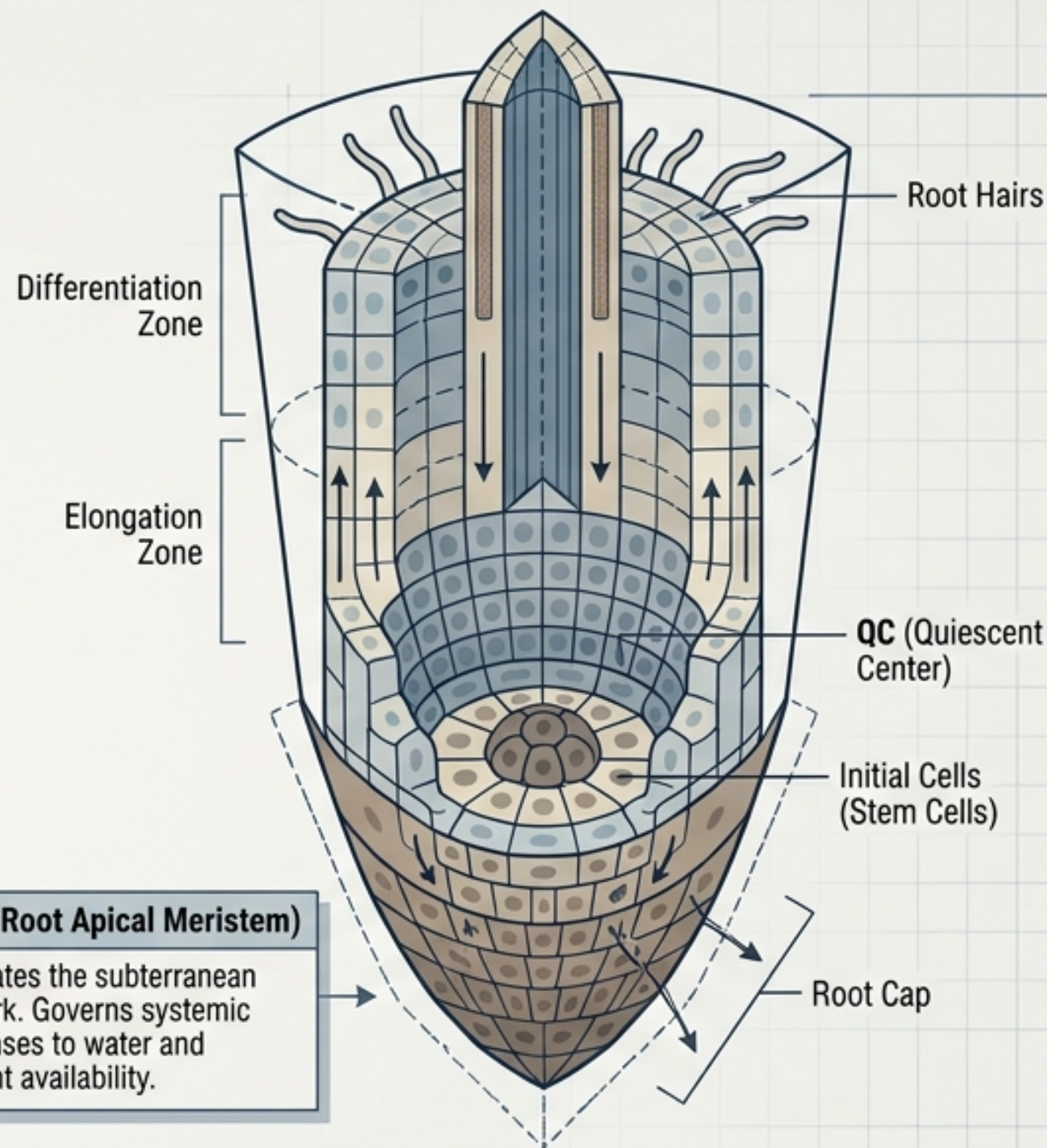
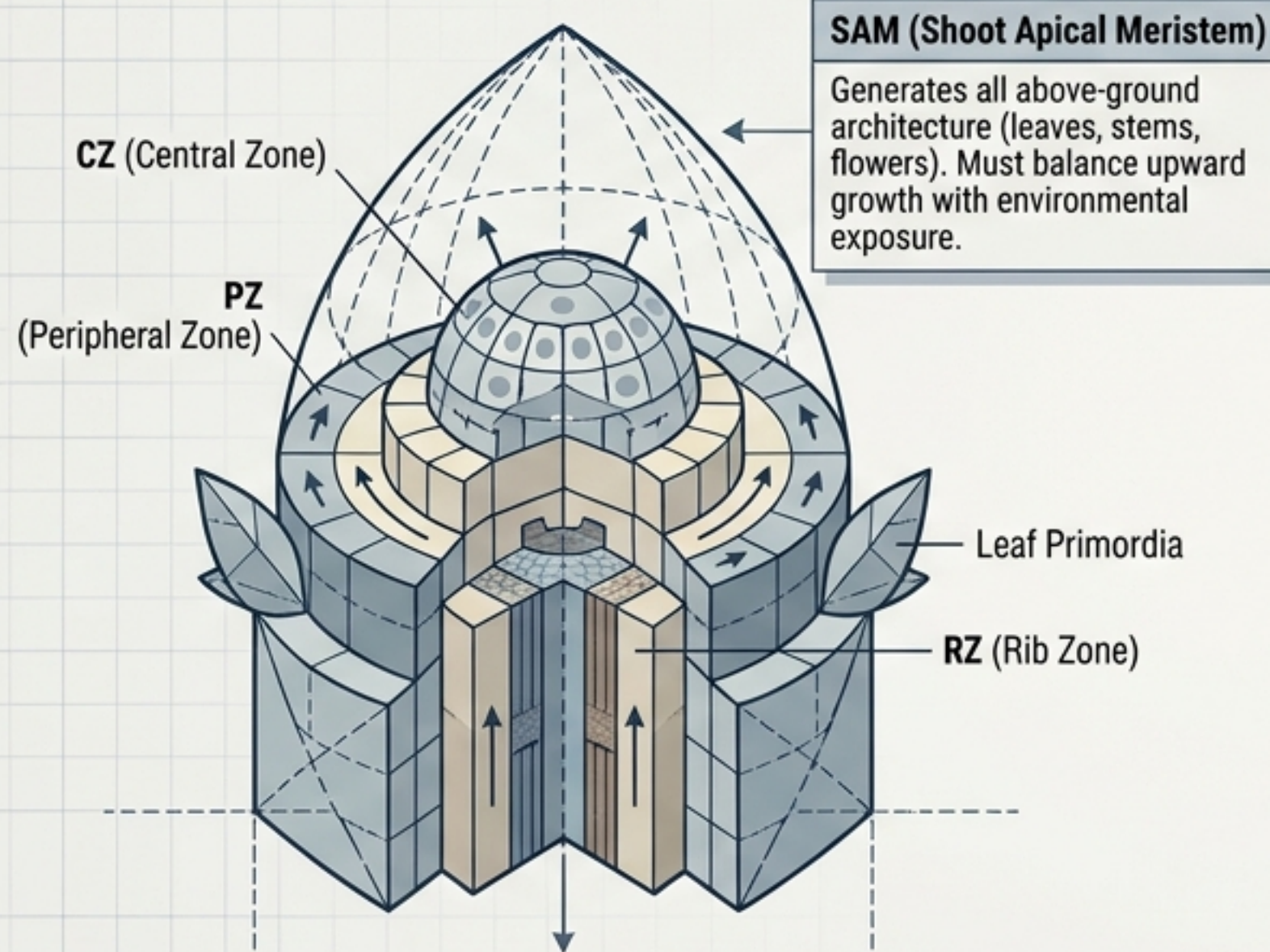
Plant Meristem Signaling & Stem Cell Maintenance
Foundations of Cellular Longevity and Plasticity in Sessile Organisms.

Context Note: Prepared for Comparative Cellular Longevity Studies.



The Meristematic Engine

Unlike animals, plants undergo continuous post-embryonic growth driven by pluripotent stem cell niches at their apices.



The Core Paradox: How do these niches maintain undifferentiated pluripotency while continuously shedding cells for specialized organogenesis, all while adapting to external stressors?

The Meristematic Engine

Central Zone (CZ)

The stem cell reservoir (Layers L1-L3). Slow-dividing, maintains pluripotency.

Peripheral Zone (PZ)

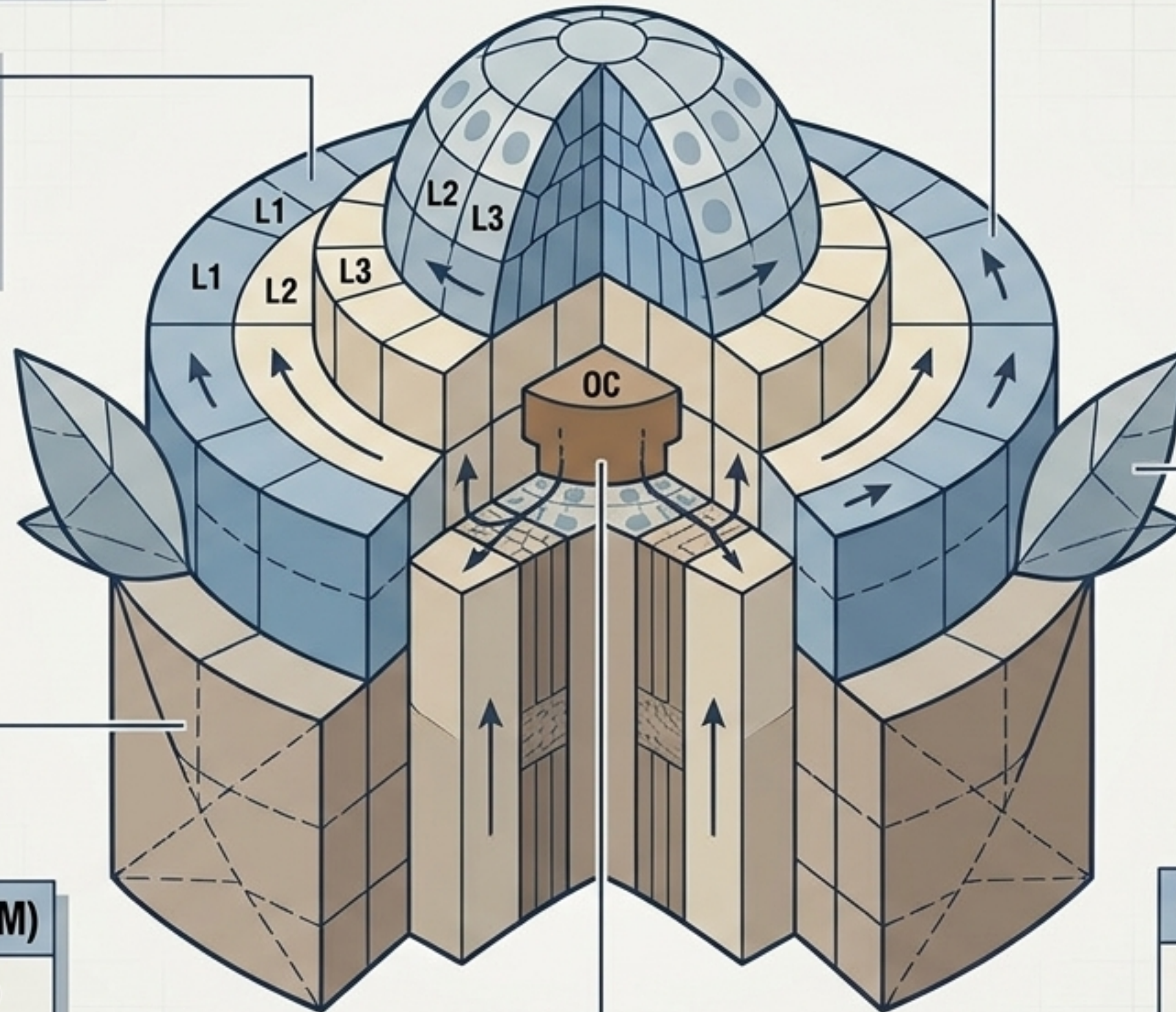
The ring surrounding the CZ. Site of rapid cell division and lateral organogenesis (leaf/axillary primordia).

Rib Zone (RZ) / Rib Meristem (RM)

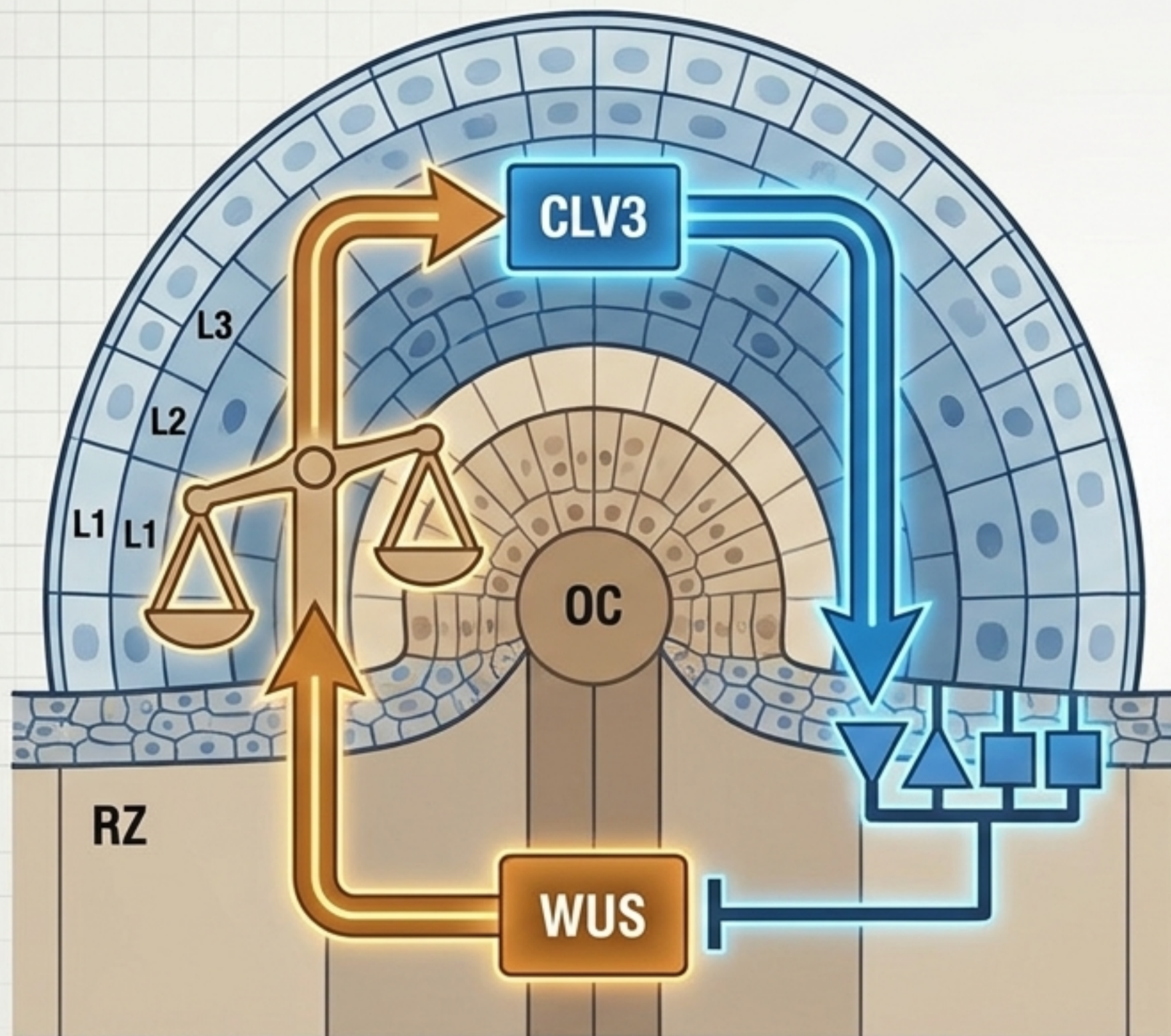
Located beneath the CZ. Divides to form the internal stem tissues.

Organizing Center (OC)

The signaling hub nested below the stem cells. Non-autonomously dictates stem cell fate above it.



The WUS-CLV3 Feedback Loop



1

Upward Activation

The homeodomain transcription factor WUS (amber) is expressed in the OC and migrates via plasmodesmata up to the CZ to maintain stem cell pluripotency and activate CLV3.

2

Downward Repression

Stem cells secrete CLV3 (electric blue), a 13-amino-acid modified peptide. CLV3 diffuses downward to bind receptor complexes (CLV1, BAM1, CLV2) in the RM.

3

The Feedback Balance

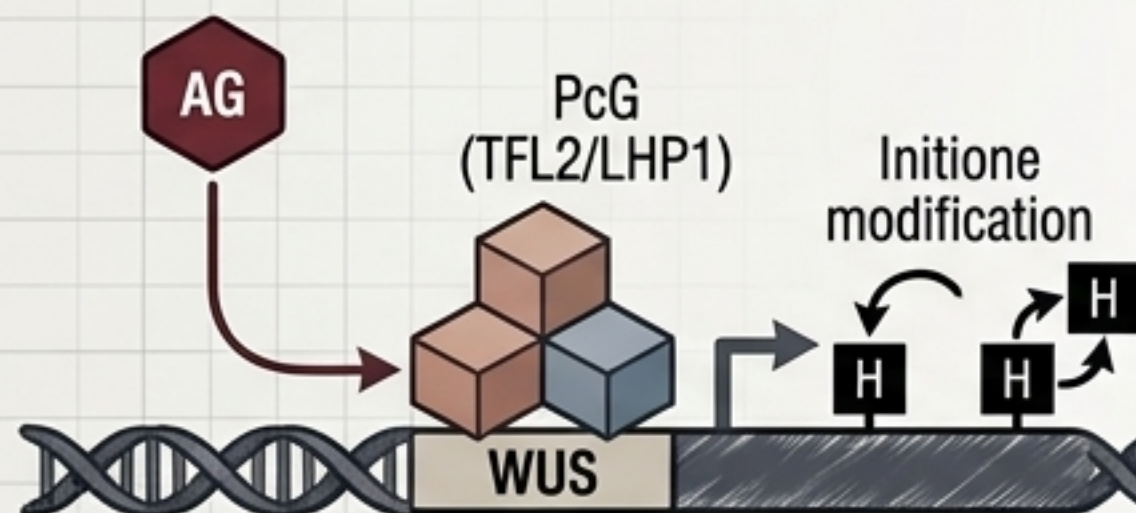
Receptor binding triggers MAPK cascades that repress WUS transcription. This limits stem cell overproliferation, maintaining a precise equilibrium between renewal and differentiation.

Floral Determinacy & Stem Cell Termination



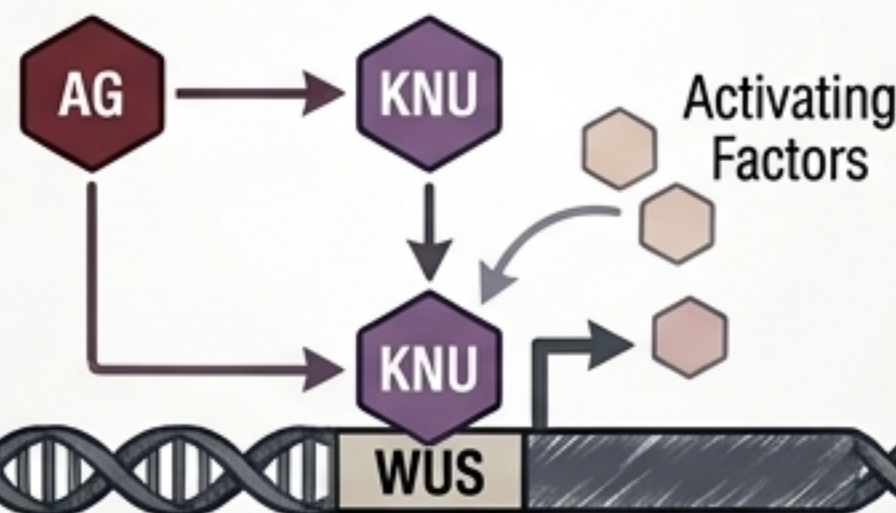
Phase 1: Early Floral Meristem (FM) Termination

AGAMOUS (AG) directly represses WUS by recruiting Polycomb Group (PcG) proteins (TFL2/LHP1). This initiates epigenetic silencing via histone modification.



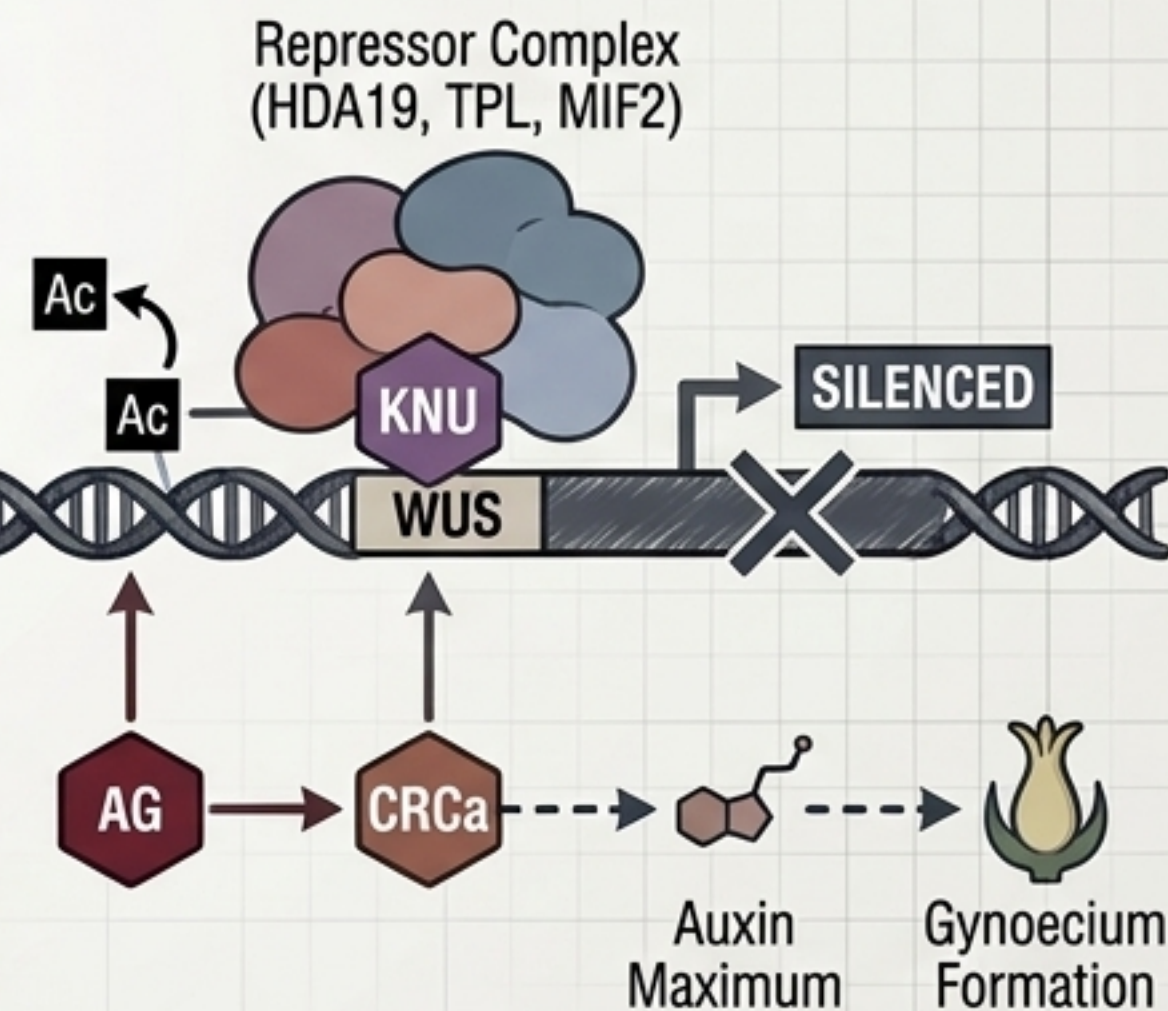
Phase 2: Stage 6 Irreversible Shutdown

AG activates KNUCKLES (KNU). KNU displaces activating factors and binds the WUS promoter.



Phase 3: The Corepressor Complex

KNU recruits a repressor complex (HDA19, TPL, MIF2) to enforce silencing through histone deacetylation. Concurrently, AG activates CRCa to establish an auxin maximum, finalizing gynoecium formation and shutting down pluripotency.



Cytokinin Crosstalk & Amplification



The ~40-Fold Amplifier

Cytokinin treatment induces WUS transcription by up to 40-fold, driving rapid meristem expansion.

CLV-Dependent Pathway

Cytokinin suppresses CLV1, removing the "brakes" from the negative feedback loop.

CLV-Independent Pathway

Direct activation via histidine kinase receptors AHK2 and AHK4.

The Localized Positive Feedback

WUS directly represses Type-A ARR (e.g., ARR5), which are negative regulators of Cytokinin. This effectively removes the dampener, amplifying Cytokinin sensitivity locally within the Organizing Center.

Type-A ARRs

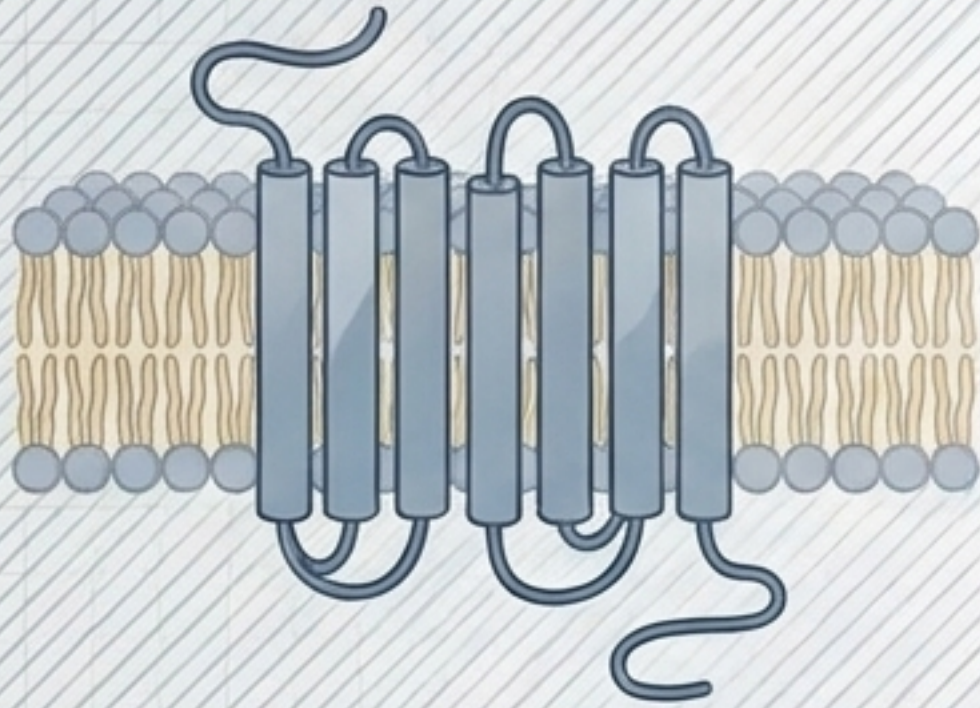
WUS



Plant-Specific Heterotrimeric G-Proteins



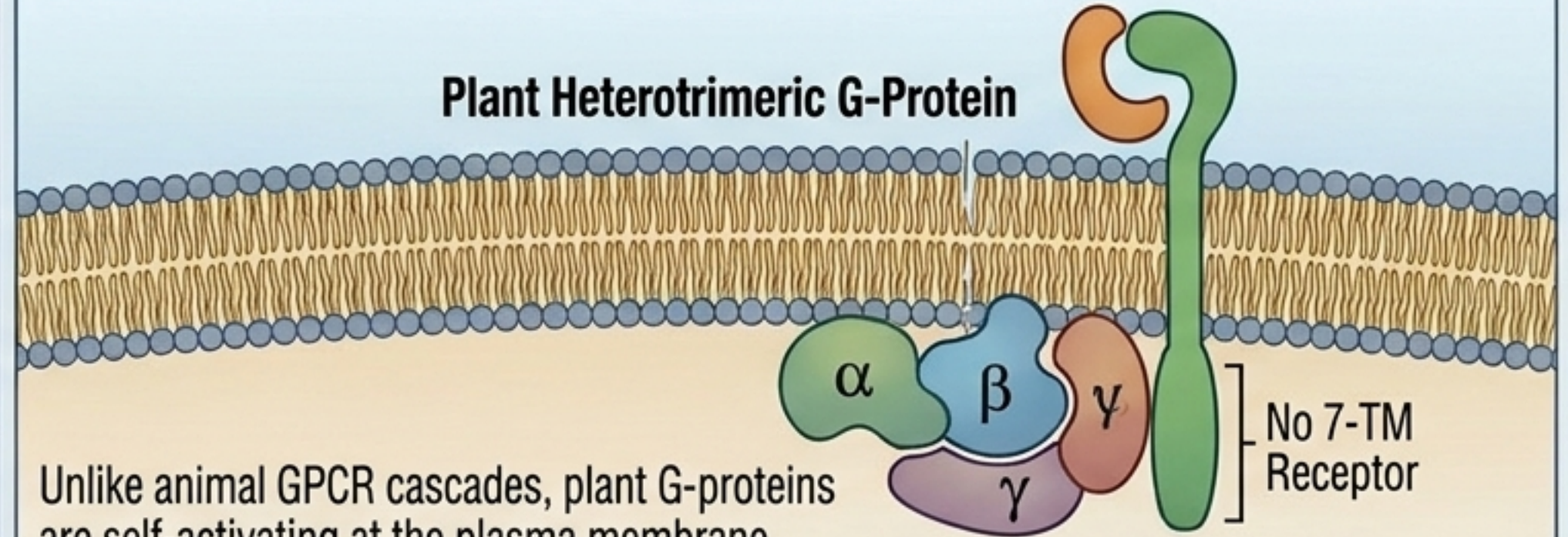
Animal GPCR Signaling



Relies on canonical 7-transmembrane receptors for activation.

The Evolutionary Divergence

Plant Heterotrimeric G-Protein



Unlike animal GPCR cascades, plant G-proteins are self-activating at the plasma membrane. They partner with Receptor-Like Kinases (RLKs) rather than canonical 7-TM receptors.

Size Calibration (Maize Model)

G-alpha subunit (CT2) binds the CLV2 ortholog FEA2 to limit SAM overproliferation.

Metabolic Lethality (Tomato Model)

Loss of the G-beta subunit (SIGB1) results in meristem lethality. Without SIGB1, a bHLH79-MYB10 dimer accumulates, triggering toxic overproduction of tyramine-derived phenolamides, terminating SAM development.

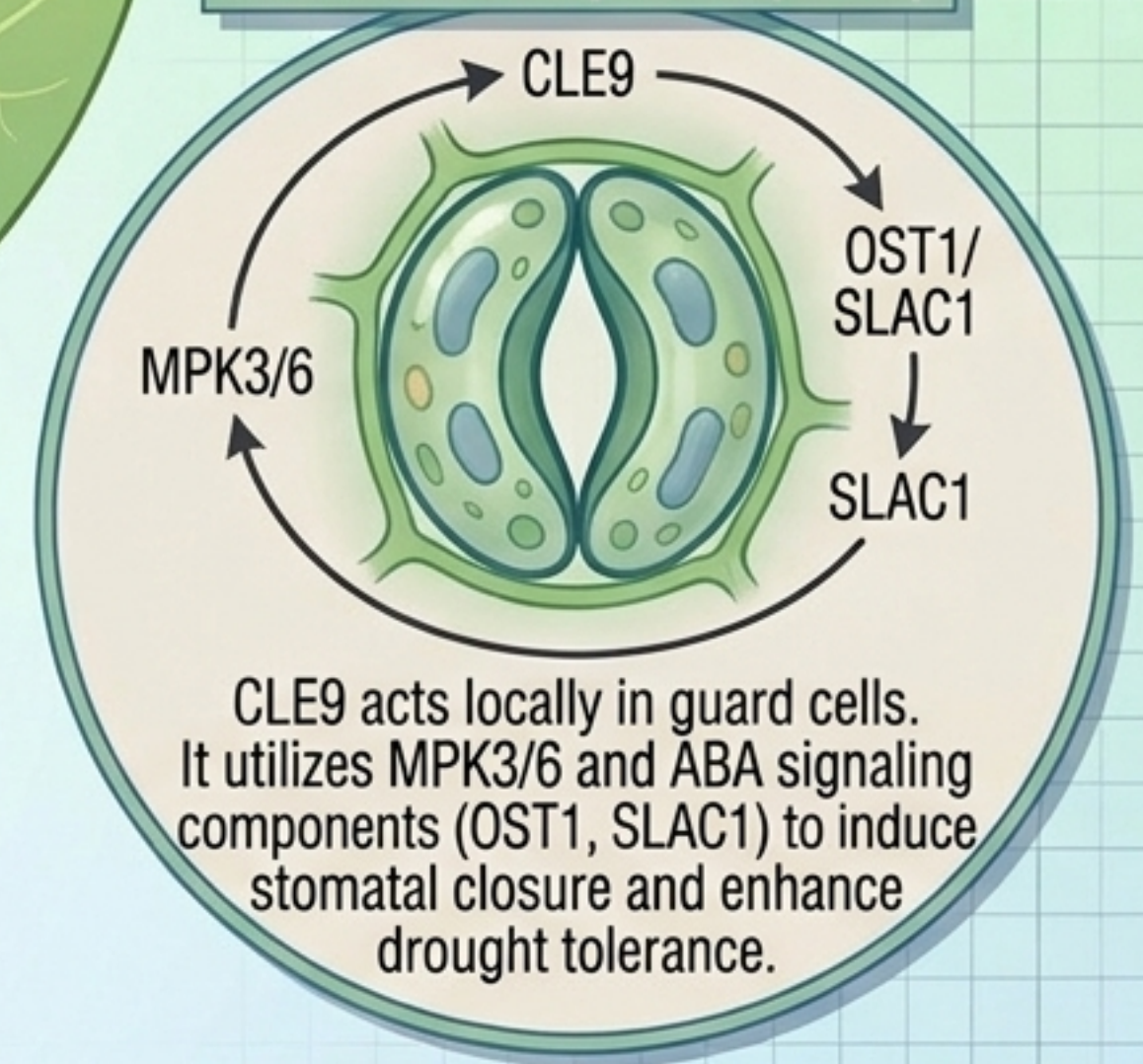
RAM Dynamics & Systemic Messengers

Systemic Response in Shoot

The CLE25 systemic peptide travels through the xylem to the shoot. It binds BAM1/BAM3 receptors, upregulating NCED3 to drive ABA accumulation and stomatal closure.

Xylem

Localized Response (CLE9)



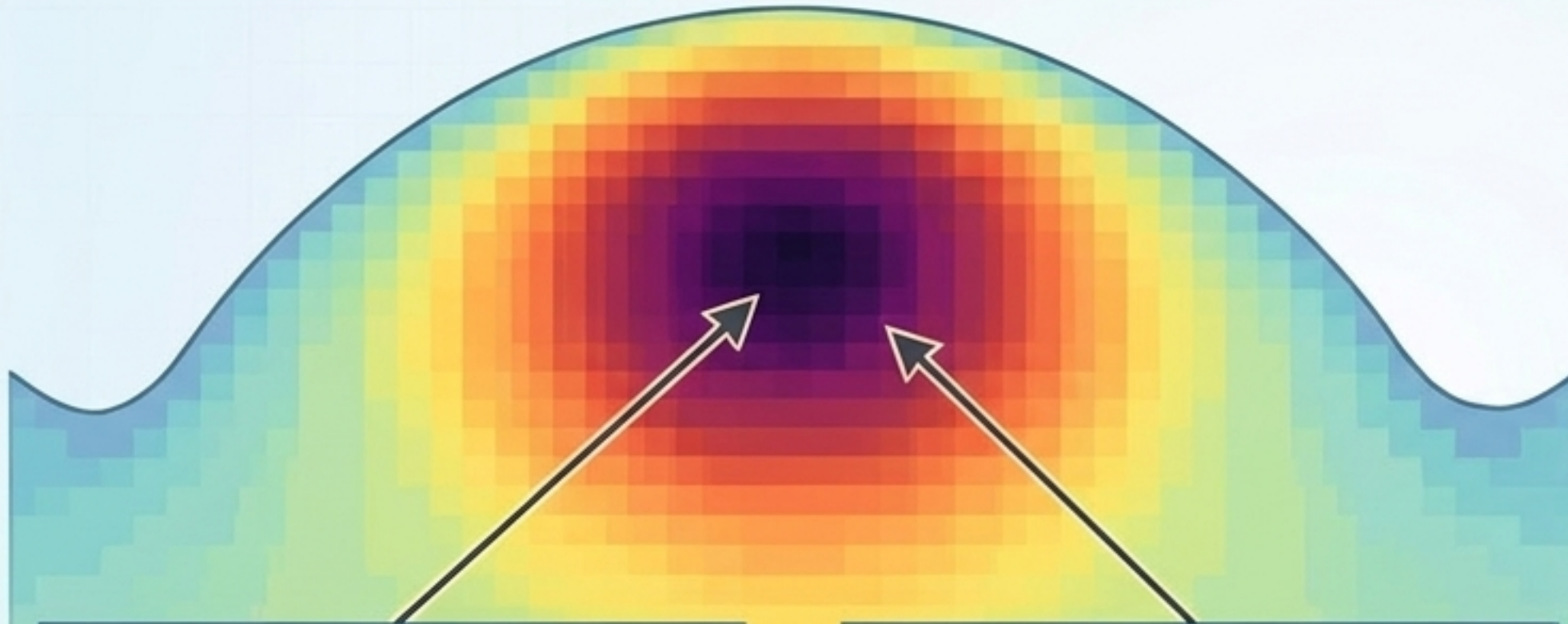
Systemic Sensing (CLE25)

Roots subjected to water deficit upregulate CLE25.

Abiotic Stress & Meristem Priming



Meristem Heatmap & Gene Expression

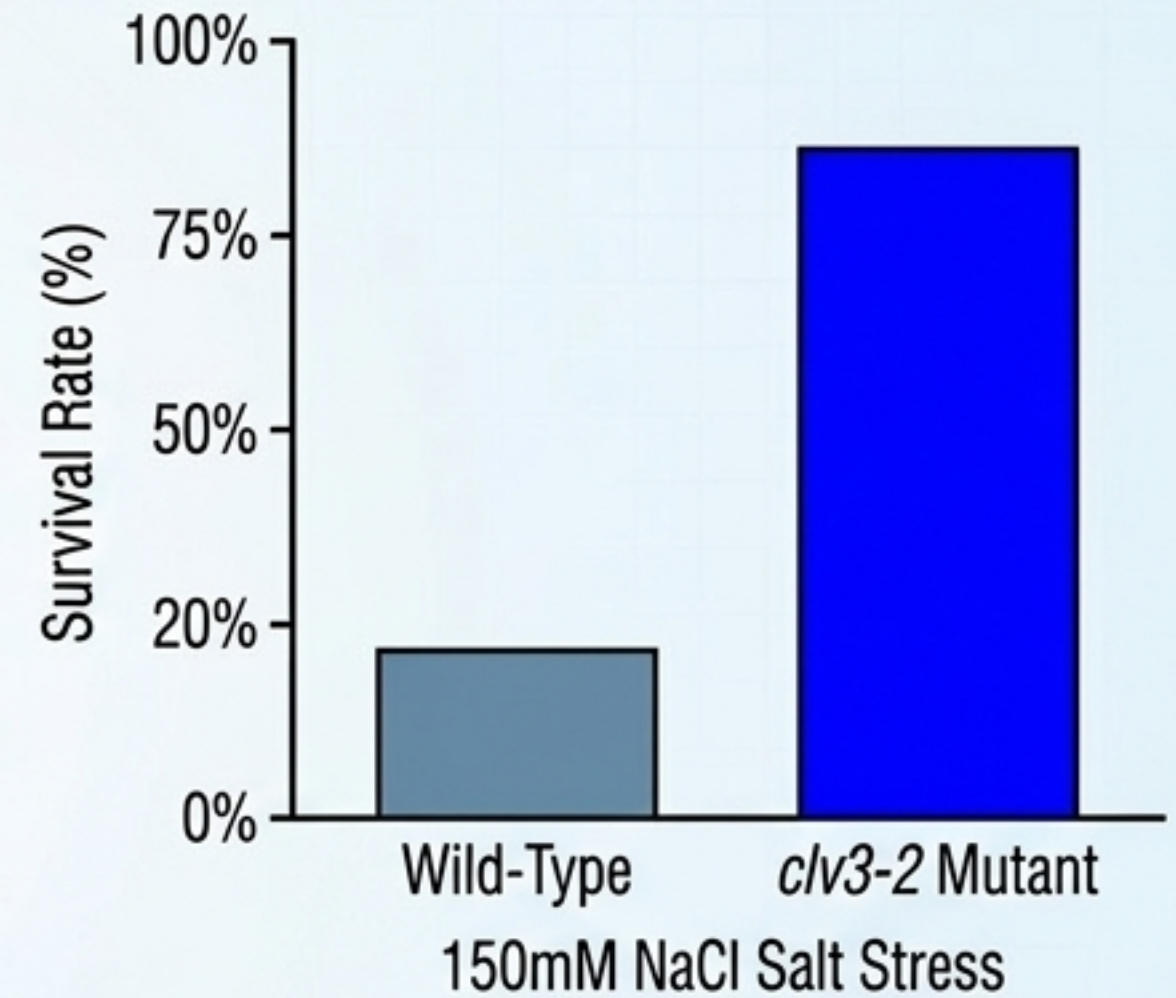


The Intrinsic Hypoxic Niche

The SAM operates in a baseline state of hypoxia, maintaining pre-accumulated stress-responsive transcription factors (NAC, WRKY, bZIP) inside the stem cell zone.

Developmental Priming Genes

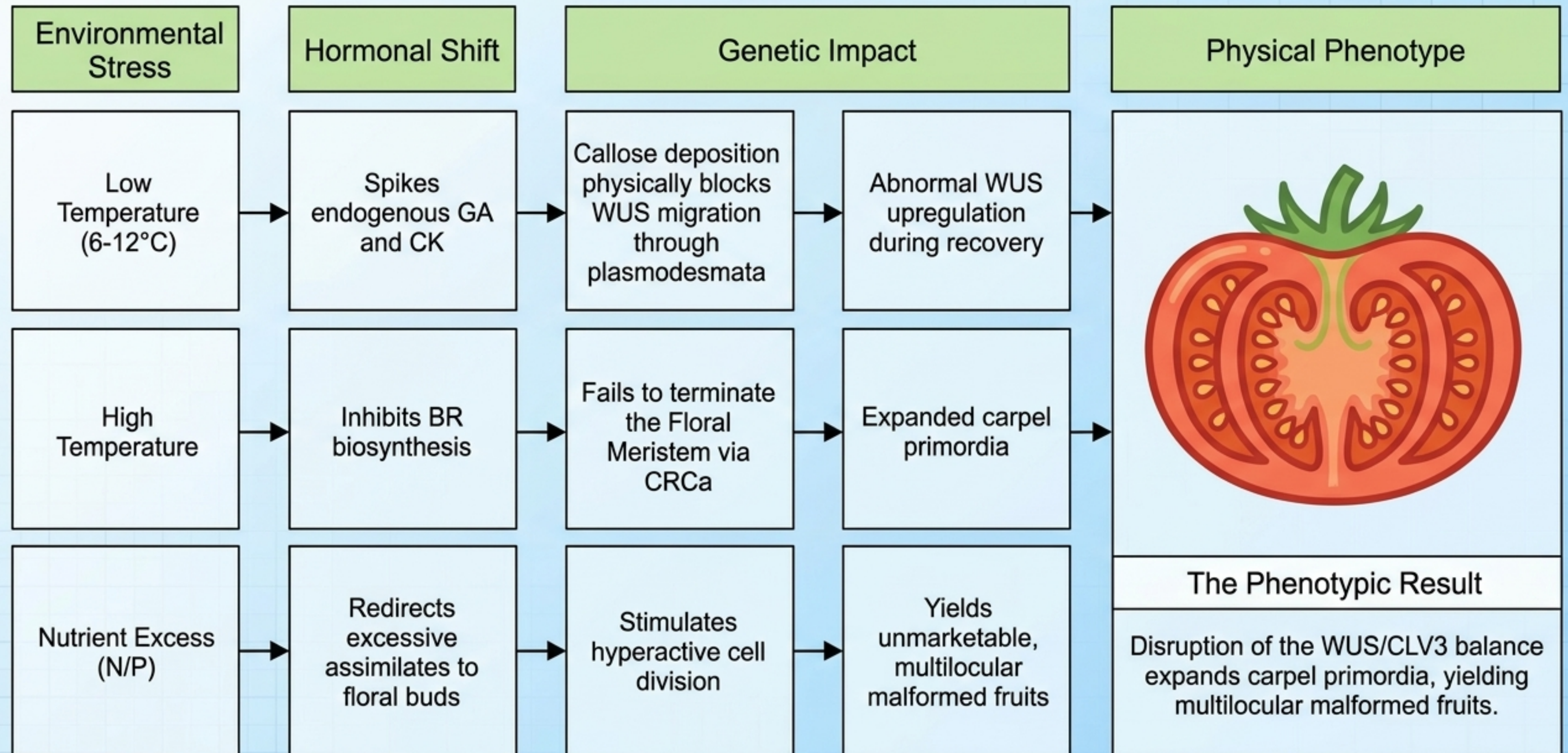
RNA-seq reveals 19 core developmental priming genes (including STM, ATHB13) that couple cell cycle regulation directly with abiotic stress resilience.



The Paradox of Mutants

Stem-cell-overproliferating mutants (*c/v3-2*, *c/v1 bam1*) exhibit vastly enhanced tolerance to severe salt stress compared to wild-type, linking meristem size directly to stress survival capacity.

Environmental Malformation (The Tomato Model)



The Peptide Defense Matrix

Abiotic Stress	Mobile Peptide	Target Receptor	Pathway Intermediates	Adaptive Response
Dehydration	CLE25	BAM1/3	NCED3 / ABA	Systemic stomatal closure
Drought (Local)	CLE9	Unknown	OST1 / SLAC1 / MPK3/6	Guard cell closure via ROS/NO
Osmotic Stress	IDA / PSK	HAE/HSL2	MKK4/5	Leaf/flower abscission to conserve resources
Salt Stress	CAPE1 / RALF22/23	FER (Internalization)	Down-regulation of salt-tolerance genes	Cell wall integrity maintenance
Nitrogen Starvation	CEP1	CEPR1/2	CEPD1/2 (descending)	Compensatory N acquisition via NRT2.1

Cross-Species Morphological Plasticity



Tomato (*Solanum lycopersicum*)

Loci: *fw2.1* (*fas*) and *fw11.3* (*lc*).

Impact: *lc* mutation increases locules by 2-4; *fas* disrupts *SlCLV3* promoter, increasing locules by up to 15. Yields extra-large fruits.



Maize (*Zea mays*)

Gene: *FEA3* (receptor).

Impact: Modulates stem cell activity via CLE peptides. Mutations produce longer ears with increased kernel rows, boosting yield.



Cucumber (*Cucumis sativus*)

Genes: QTLs *In1.1* and *In1.3*.

Impact: Drives multilocular traits, expanding fruit width and altering flavor profiles.

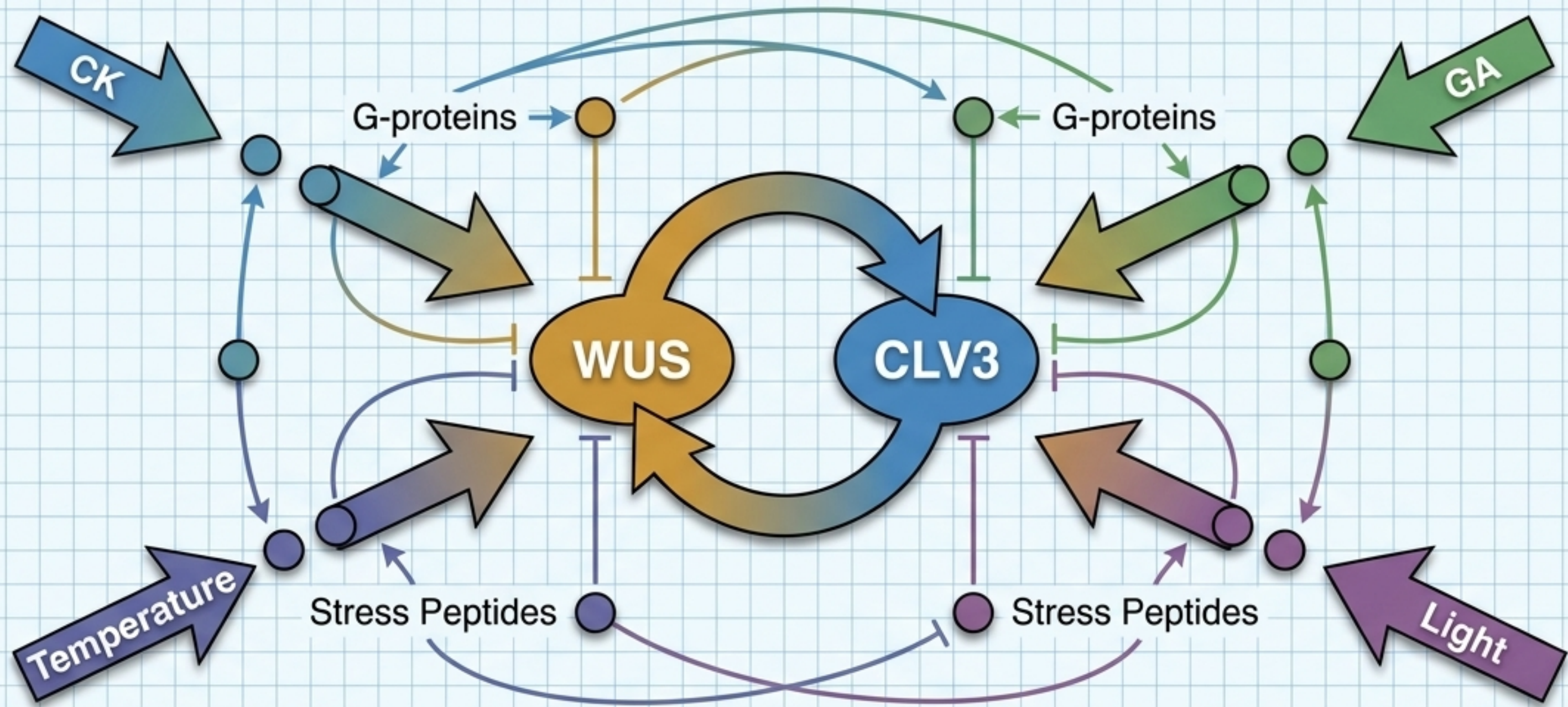


Rapeseed (*Brassica napus*)

Gene: *Bra034340* (*CLV3* homolog).

Impact: Regulates silique width. Multilocular varieties possess higher seed counts and enhanced disease resistance.

SYNTHESIS: The Integrated Meristem Network



The Computational Hub

The SAM uses unified circuitry to weigh the mandate for continuous growth against environmental trauma.

Core Rheostat

CLV-WUS maintains the baseline stem cell equilibrium.

Hormonal Amplifiers

Cytokinin and Auxin act as logic gates for expansion or organogenesis.

Environmental Integrators

G-proteins and systemic peptides directly tune meristematic volume and stress resilience in real-time.

Future Horizons: From Plant Plasticity to Cellular Aging

Unlike animals, plants achieve extreme lifespans through the flawless, indefinite maintenance of pluripotent niches exposed to constant environmental trauma.

Key Questions for Module 2:

- Can the epigenetic resilience (e.g., Polycomb-mediated silencing) of plant stem cells inform mechanisms of human cellular senescence?
- Does the 'stress-priming' of the SAM—coupling hypoxia and active cell division—offer novel models for extending pluripotency in mammalian tissue repair?

Next: Comparative Frameworks in Mammalian Stem Cell Maintenance.

